

Lab Write-up: Setting up my Hybrid Cloud & Zero-Trust Lab (Architecture Diagram Included)

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Focus: Cloud Networking | Identity | Scalable Infrastructure

What I was trying to do:

I wanted to stop just reading about Azure and actually build a hybrid environment that mimics a real-world business setup. My goal was to bridge my home lab (Hyper-V) with an Azure VNet using a Site-to-Site VPN and then lock everything down using a Zero-Trust framework.

The Setup:

Instead of a flat network, I built a Segmented VNet architecture. While it's a single VNet for now, I've subnetted it so it's "Hub-and-Spoke ready." The IP schema is non-overlapping, and the identity/gateway services are already isolated, so I can easily peer this out to more VNets as I scale.

- **Infrastructure:** I set up core subnets for Identity, Web, and DB. I gave the workloadSubnet and vdiSubnet full /24 blocks to give them room to grow, while keeping the specialized service subnets tight at /27.
- **The Legacy Tier (On-Prem):** On the Hyper-V side, I maintained a legacy environment including a combined App/DB VM and a Test Client. This represents the soon-to-be retired servers that I am systematically refactoring into a modern, tiered architecture in Azure.
- **Tunnel:** I established an S2S VPN tunnel (IPsec/IKEv2) from my home network to the GatewaySubnet. Getting the parameters to match was a headache at first, but once it connected, the replication between my local Hyper-V host and the Azure AZ-DC-01 was solid.
- **Modern Management:** I moved away from "RDP-everywhere" by deploying AZ-WAC-01 (Windows Admin Center). To keep it secure, I restricted its management interface

so it's only accessible from the AZ-jumpbox, creating a controlled "Admin-Only" management path within the workload tier.

Identity & Security (Zero-Trust):

Since I'm focusing on the Identity side, I used AZ-ADSYNC-01 to bridge my local AD to Entra ID. This let me play with the "Identity is the new perimeter" concept:

- Conditional Access: I set up a policy so that if I'm not on my home VPN (my "Trusted Location"), I get prompted for MFA immediately.
- Intune & Compliance: I linked my test devices to Intune. If a device isn't compliant (e.g., BitLocker is off), Entra ID blocks it from hitting my private-linked Azure Files shares.
- VDI Scalability: I deployed AZ-VDI-SS as a Virtual Machine Scale Set. This simulates a remote workforce where desktops scale with demand, using FSLogix profiles stored on Private Endpoints to keep user data persistent across stateless VMs.

Troubleshooting & Lessons Learned:

The Routing Gap: I initially couldn't get my local VMs to talk to Azure. It ended up being a mix of an improperly sized GatewaySubnet and a missing NSG rule.

- Firewall Layers: I learned that an "Allowed" NSG rule doesn't mean much if the Windows Guest Firewall is still dropping ICMP. It taught me to troubleshoot the entire path, not just the cloud settings.
- Micro-segmentation: Putting an NSG "Shield" on every single subnet border was tedious but worth it. It's the only way to truly enforce a "Default Deny" posture and stop lateral movement.

What I learned:

Networking in the cloud is easy until it isn't. This lab helped me really wrap my head around how User Defined Routes work and why segmenting your traffic by "role" is better than just dumping everything into one big subnet. It's a foundation that's "Small Business" today but "Enterprise Ready" tomorrow.

Future Roadmap:

- Scale to Hub-and-Spoke: Transition the current segmented VNet into a formal Hub-and-Spoke model by peering out the VDI and Web tiers into their own dedicated Spokes.
- Centralized Security: Implement Azure Firewall or a third-party NVA (Network Virtual Appliance) to move from decentralized NSGs to a centralized traffic inspection model.
- Infrastructure as Code (IaC): Use Bicep or Terraform to automate the deployment of this entire stack, making the "Hybrid Lab" repeatable and version-controlled.

